NAEOS Whitepaper

Version 3.7.1

NAEOS Foundation

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NAEOS — Official Whitepaper

Nusantara AI Engineering Operating System

"Architecture Drives Engineering."

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Executive Summary

NAEOS is an open-source declarative engineering platform that transforms specifications into high-quality software systems through a consistent, validated, and extensible pipeline. NAEOS is not just a project generator — it is an *engineering runtime* that understands specifications, builds an internal model (NEIR), orchestrates execution plans, generates artifacts, validates results, and keeps projects aligned with their specifications throughout the entire lifecycle.

Under the motto **"Architecture Drives Engineering."**, NAEOS enables organizations to describe their system **once**, then derive code, documentation, configuration, and AI context from a shared engineering model across multiple languages and frameworks.

The current repository state includes Specification Language v2, a multi-adapter AI compiler, a NEIR-aware LSP, constitution-based governance, a marketplace, a production server daemon (`naeos serve`), SBOM and artifact signing, Helm chart scaffolding, and compliance-oriented policy templates for SOC 2, HIPAA, and GDPR workflows.

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1. Background & Problem Statement

1.1 The Fragmentation of Software Engineering

The software industry faces a systemic fragmentation crisis:

- **Multi-language, multi-framework** — A single system now spans Go, TypeScript, Python, Java, and Rust simultaneously, each with its own frameworks, conventions, and toolchains.

- **Specification–implementation drift** — Documentation and code diverge over time; no automated mechanism guarantees alignment.
- **Loss of engineering context** — Architectural decisions, ADRs, and organizational knowledge live only in individuals' heads — undocumented and untraceable.
- **AI tooling explosion** — Each AI coding agent (GitHub Copilot, Claude Code, Cursor, Gemini CLI, Codex, OpenCode) uses a different context and instruction format, forcing organizations to maintain many configuration files with identical content but different formats.
- **Inconsistent governance** — Without enforcement mechanisms, policies, standards, and organizational rules remain static documents that are never executed.

1.2 The Cost of Inconsistency

The impact is directly measurable: rework, expensive audits, painful migrations, knowledge loss when team members leave, and failure to meet regulatory compliance (SOC 2, HIPAA, GDPR) due to a lack of verifiable audit evidence.

1.3 Thesis

> **The specification is the single source of truth. Everything — code, documentation, configuration, AI context, deployment artifacts — must be derived from the specification through a deterministic, validated, auditable pipeline.**

NAEOS was built to prove this thesis.

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2. Vision & Mission

Vision

Build an open-source engineering platform that enables developers and organizations to **describe their system once**, then build, validate, and evolve software across multiple languages, frameworks, and platforms — with an engineering constitution enforced automatically.

Mission

1. Make the declarative specification the single source of truth for every engineering artifact.
2. Provide a deterministic, validated, auditable compilation pipeline.
3. Bridge the gap between governance, specification, and execution through executable policies.
4. Build a universal bridge to all AI coding agents without vendor lock-in.
5. Deliver **technological sovereignty**: a vendor-neutral, language-neutral, cloud-neutral platform.

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3. Foundational Principles: The Engineering Constitution

NAEOS formalizes its principles in the **Engineering Constitution** (NAEOS-CON-001) — the highest normative document, acting as the source of executable rules. Its normative hierarchy:

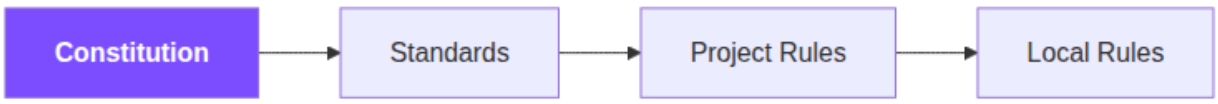


Figure 1: mermaid diagram 1

The twelve constitutional articles:

#	Article	Essence
I	**Specification First**	No specification = no implementation
II	**Knowledge Preservation**	Every engineering decision must be documented (ADR, RFC, API contract)
III	**Traceability**	Requirement → Spec → Architecture → Code → Test → Deployment
IV	**Single Source of Truth**	No two official artifacts may state conflicting normative information
V	**Human Accountability**	AI assists; humans decide and are accountable for releases
VI	**Security by Design**	Security is part of design from the start, not a final stage
VII	**Documentation as Code**	Documentation is versioned, reviewed, validated, and compiled
VIII	**Reproducibility**	Identical input produces identical output
IX	**Vendor Neutrality**	No dependency on a single AI vendor
X	**Extensibility**	Extension without modifying core specifications
XI	**Quality Before Velocity**	Speed must not sacrifice security, maintainability, correctness
XII	**Continuous Improvement**	The constitution evolves through RFC and ADR processes

What makes NAEOS distinctive: **the constitution is not merely a document** — its articles are compiled into executable rules enforced by the Rule Engine, executed by the Validator and Compiler, and checked by AI Review:

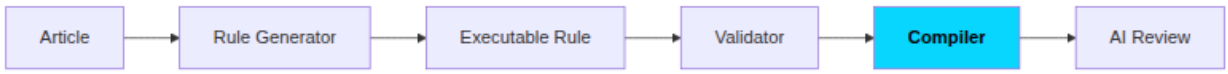


Figure 2: mermaid diagram 2

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4. Platform Architecture

4.1 Layered Model

NAEOS connects five principal layers:

4.2 Compilation Pipeline

The core pipeline follows a deterministic flow:

The pipeline is backed by production-grade infrastructure:

- **Stage caching v2** — per-stage cache keyed by NEIR SHA-256 hash; hit rate inspectable via `--profile`.
- **Parallel generation** — concurrent multi-adapter execution (±1.4ms for 3 adapters vs ±3ms sequential).
- **Profiling & memory profiler** — per-stage timing/memory, heap diffing, leak detection.
- **Pipeline middleware** — composable chain (log, metrics, auth, cache).
- **Event sourcing & observability** — execution snapshots, telemetry tracing, WebSocket live updates.

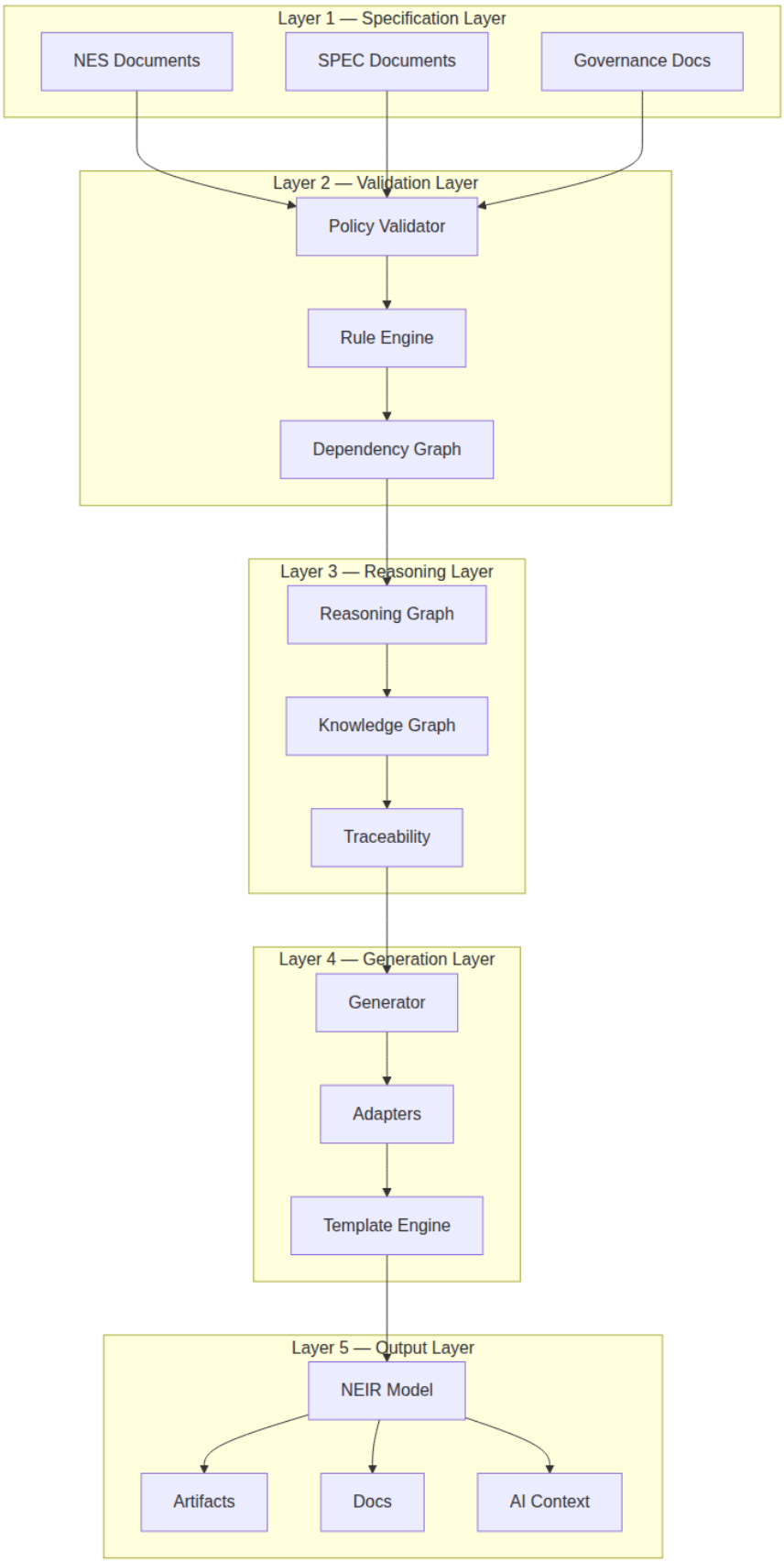


Figure 3: mermaid diagram 3

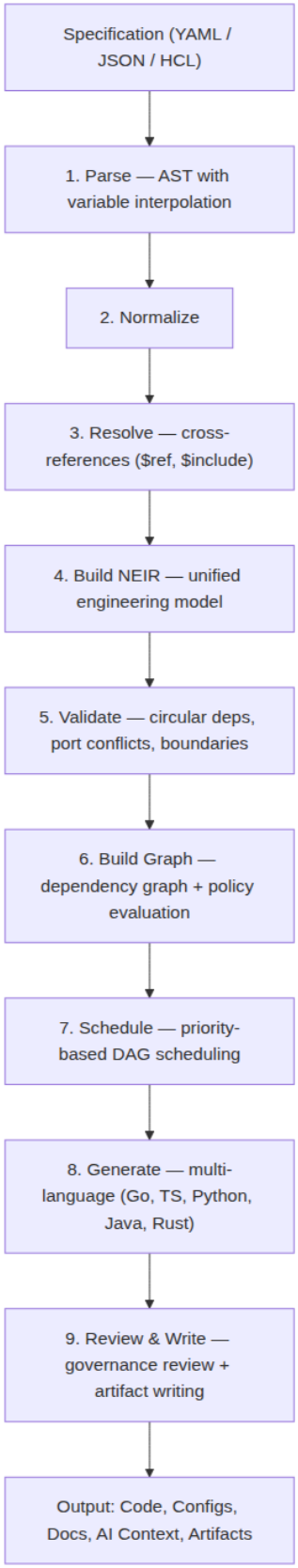


Figure 4: mermaid diagram 4

4.3 End-to-End Data Flow

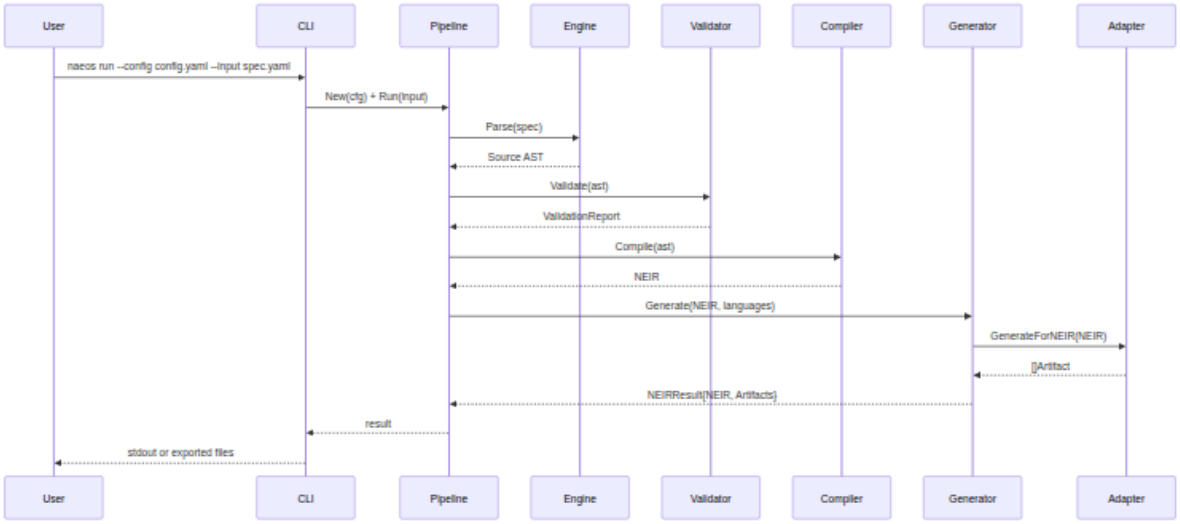


Figure 5: mermaid diagram 5

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5. Core Components

5.1 Specification Language v2

The NAEOS specification language is the single source of truth — human-readable and machine-validatable:

Feature	Syntax	Purpose
Variable interpolation	`${var}`	Reference values within a spec
Environment variables	`$env{VAR}`	Resolve from environment
Cross-references	`$ref{path}`	Reference between spec sections
File composition	`$include{file}`	Multi-file specs
Built-in functions	`$fn{upper/slug/default/...}`	Value transformations
Conditional sections	`$if{cond}...$endif`	Conditional content

Version 2.0 adds **conditional modules** (`Condition` field), **environment profiles** (`ActiveProfile`/`Inherits`), and schema-based validation with automatic minimum-version checking.

5.2 NEIR — NAEOS Engineering Intermediate Representation

NEIR is the **unified engineering model** representing the entire system. NEIR is not merely an AST — it is a canonical model spanning 14 domains:

`project, architecture, domain, module, component, service, API, storage, infrastructure, security, AI, documentation, deployment, testing, metadata`

Technical characteristics:

- **Lazy loading** — per-section accessors; only required data is loaded.
- **Schema versioning** — semver schema registry with remote validation (`naeos schema validate`).
- **Deterministic** — identical input produces identical NEIR (Article VIII).
- **Diff-able** — structural NEIR comparison to track system evolution.

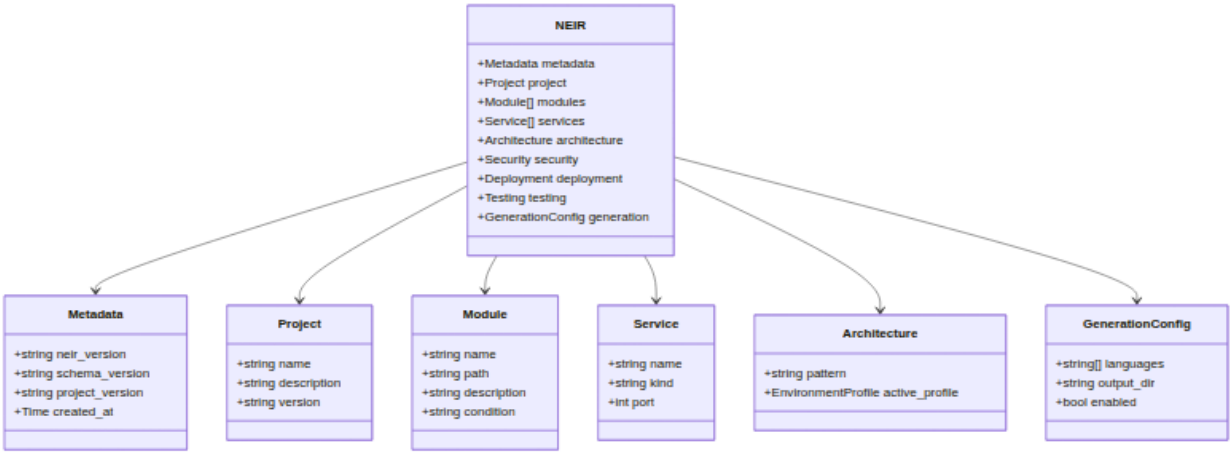


Figure 6: mermaid diagram 6

5.3 Multi-Language Generator

Language	Supported Stack	Status
Go	—	Active
TypeScript	—	Active
Python	—	Active
Java	JUnit 5	Active
Rust	Axum 0.7	Active

Each adapter produces consistent artifacts: code, configuration, Dockerfiles (5 languages), docker-compose, and Kubernetes manifests.

5.4 Kernel & Runtime

- **Service Registry** — centralized service registration
- **Event Bus** — internal pub/sub with PipelineObserver
- **Telemetry** — spans, batched export, HTTP exporter, Prometheus metrics
- **Lifecycle Management** — health checks, graceful shutdown, WebSocket draining

5.5 AI Integration & Compiler

The NAEOS compiler transforms NEIR into **AI instruction sets** for 6 target tools:

Additionally:

- **MCP Server** — Model Context Protocol for agent integration (validate_spec, compile_spec, list_artifacts, get_pipeline_status, export_terraform, list_plugins).
- **Context Bundles** — LLM-optimized project summaries enriched with dependency graphs, security context, and cloud resource mapping.
- **AI Compiler Adapter** — streaming spec compilation to LLMs (OpenAI, Anthropic, Ollama) with true SSE streaming.
- **Prompt Library** — centralized YAML-based prompt templates (LLM + compiler adapters) with custom template functions.
- **NEIR-aware LSP** — Language Server Protocol for spec YAML: autocomplete, diagnostics, hover, go-to-definition, code actions — with real parser integration.
- **VS Code extension** — extension generator (`naeos dx vscode-gen`) with TextMate grammar and LSP client.
- **AI Constitution** (NAEOS-CON-002) — articles specifically governing AI's role in engineering, aligned with Articles V and IX.

5.6 Governance & Compliance

Policy & Governance: - Policy Evaluator — 9 operators, 5 built-in rules - Artifact Review — artifact

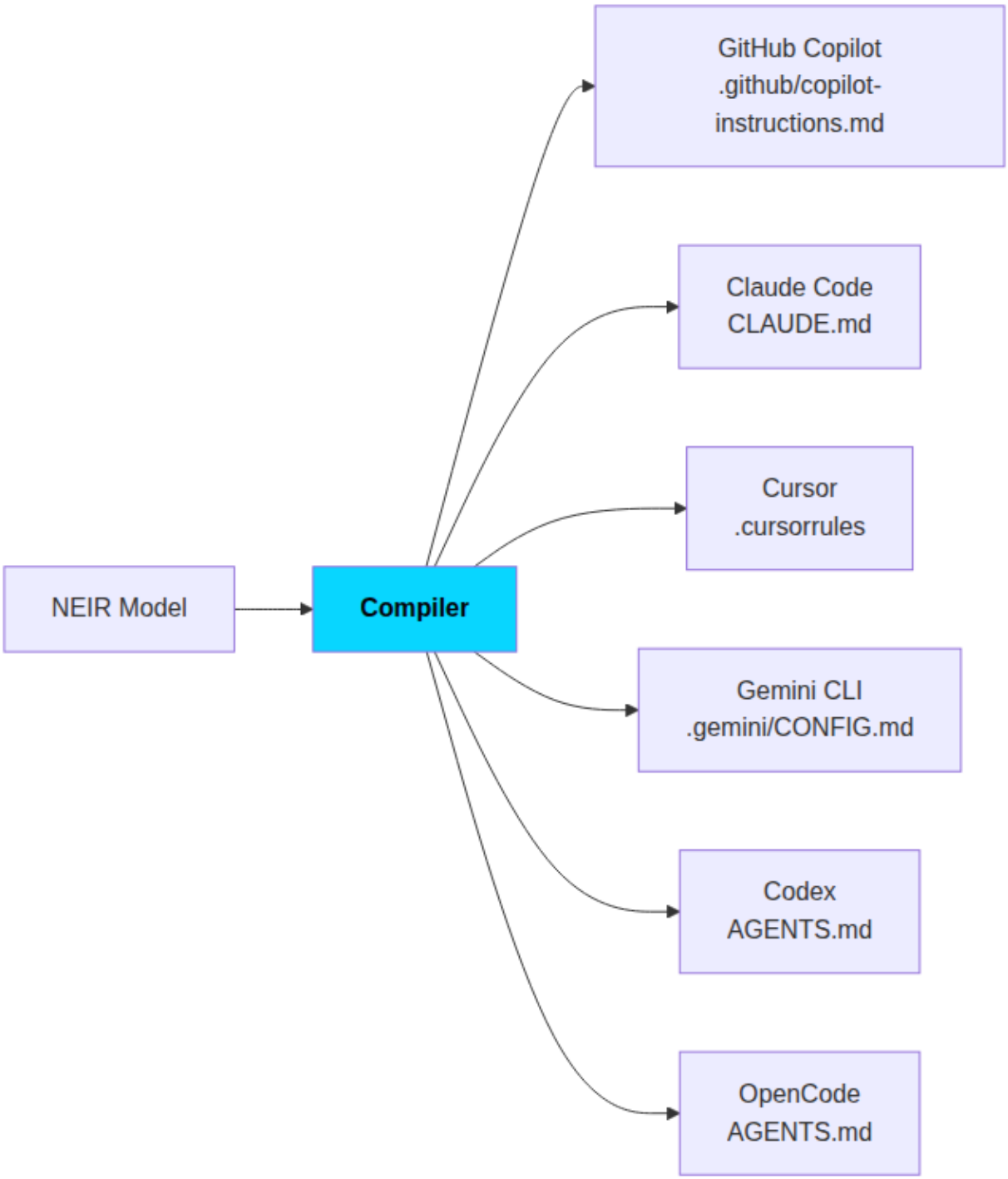


Figure 7: mermaid diagram 7

5.7 Ecosystem Marketplace

Marketplace	Function	Content
Profile	Publish, search, download	5 built-in industry profiles: SaaS, AI Agent, FinTech, Healthcare, Government
Plugin	Install/uninstall/search	WASM runtime (wazero), hot-reload, event bus, public registry
Template	Publish starter projects	Scaffolding with CI/CD, SDK, and WASM entry point

Plugins execute safely through a **JSON-over-stdin/stdout sandbox** and **WASI**, with signature verification and lazy loading.

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6. Differentiation: NAEOS vs Conventional Approaches

Dimension	Conventional Approach	NAEOS
Source of truth	Many (code, docs, wiki, chat)	One: declarative specification
Code & spec	Drift over time	Derived together from spec, always aligned
AI context	Manual per-tool files, easily stale	Compiled from NEIR for 6 tools at once
Governance	Static documents, never executed	Constitution → Rule Engine → Validator, enforced automatically
Traceability	Manual, incomplete	Automatic: requirement → deployment
Compliance	Manual, expensive audits	Automatic reports (SOC 2/HIPAA/GDPR) + verifiable audit chain
Extensibility	Fork or separate tooling	Official plugin WASM, profile, template marketplace

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7. Release History & Roadmap

Releases achieved

- **v0.x** — Foundation: parser, NEIR, pipeline, CLI, 6-adapter compiler, cloud (AWS/GCP/Azure), AI integration, distributed task execution, event sourcing
- **v1.x** — Stability: database layer (PostgreSQL/MySQL/SQLite), 999 lint issues resolved, production hardening, prompt library, observability dashboard
- **v2.x** — Platform: Supabase integration, NEIR v2.0 (conditional modules, env profiles), hierarchical RBAC, OAuth2/OIDC, SSO (SAML 2.0, LDAP), compliance frameworks, hashed + encrypted audit chains, stage caching, LSP, VS Code extension, real distributed builds, pipeline/memory profiling
- **v3.0.0** — Ecosystem release: 20+ new features, changelog, migration guide, deprecation notices
- **v3.1.0** — Performance release: pipeline caching on `naeos run`, run-level profiling (`--profile`/`--pprof`), architecture patterns (monolithic/microservices/serverless), WASM plugin hardening
- **v3.2.0** — Operations release: production server daemon (`naeos serve`) with TLS, graceful shutdown, and systemd integration; policy registry + control plane, runtime execution gateway, immutable evidence store, independent verification, MCP resources/prompts/completions/ping
- **v3.3.0** — Supply-chain release: SBOM generation (CycloneDX), Ed25519 artifact signing, SBOM verifier
- **v3.5.0** — Observability & control-plane release: OTLP tracing, SLO alerting, SIEM export (CEF/NDJSON), and the investor demo control plane
- **v3.6.0** — Unified control plane release: reusable authorization, approvals, and tamper-evident evidence across the investor demo, coupled with supply-chain hardening (gitleaks, NOTICE/SBOM in releases, DCO enforcement)
- **v3.4.0** — Deployment release: Helm chart scaffolding, air-gapped bundles, config providers (env/file/K8s secret/Vault)

Platform health metrics (current)

Metric	Value
Test coverage	~87% (target 85%)
Lint pass rate	100% (linters, incl. gosec & errorlint)
CLI commands	200+ (284 CLI doc pages)
CLI test coverage	~80.8% (target 100%)
Packages 80% coverage	13+ (watch, rollback, cicd, distributed, gateway, websocket, configschema, monitoring, configreload, database, auth, supabase, and more)

Current execution strategy

The project is sequencing work around proof and adoption rather than opening another feature-expansion cycle:

1. **P0 — Public consistency:** synchronize website, SEO, whitepaper, FAQ, About, and public documentation.
2. **P1 — Golden Path:** make P1.6–P1.10 the primary reproducible developer proof, with P1.11 providing independent verification.
3. **P2 — External adoption:** work with the first 5–10 developers or repositories and capture evidence, friction, and failure modes.
4. **P3 — Ecosystem:** prioritize SDKs, integrations, and marketplace work from recurring adoption needs.
5. **P4 — Commercialization:** evaluate Cloud/Enterprise packaging after technical adoption evidence exists.

These are execution stages, not software release versions. See ROADMAP.md for the current public execution roadmap.

8. Licensing & Project Governance

- **License**: Apache License 2.0 — free to use, modify, and distribute; commercial and internal use permitted with attribution.
- **Document governance**: all standards follow the NES (NAEOS Engineering Specification) flow, ADR (Architecture Decision Record), and RFC with mandatory review.
- **CI/CD**: every PR must pass lint + test + coverage checks; coverage drops block merges; every new API/feature must include documentation before merge.
- **Releases**: GoReleaser multi-platform builds (linux/darwin/windows × amd64/arm64) + multi-arch Docker image + automated blog post (EN/ID) per release.

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9. Use Cases

Scenario	Value Delivered
Multi-language startup	One spec produces Go + TypeScript code and infra, removing ~70% of boilerplate
Regulated organizations (fintech/healthcare/gov)	Automated policy enforcement + SOC 2/HIPAA/GDPR reports + tamper-evident audit chain
Teams adopting AI coding agents	AI context compiled for 6 tools from one source — no more stale files
Multi-cloud enterprises	Terraform HCL for AWS/GCP/Azure from NEIR; industry profiles standardize architecture
Large evolving platforms	Structural NEIR diff, migration engine v0.1→v0.3, rollback, and spec repair
Teams enforcing quality	Comprehensive validation (circular deps, port conflicts), fuzz testing, standardized benchmarks

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10. Risks & Adoption Considerations

- **Plugin ecosystem maturity** — Currently 0 community plugins; targets of 5+ (Q1 2027) and 20+ (Q3 2027). Mitigation: plugin SDK, template generator, and public registry are already available.
- **Spec language learning curve** — Mitigated by LSP, TUI wizard, and 57 NES specification documents.
- **AI determinism challenges** — Constitution Articles V and VIII ensure AI only assists within a deterministic pipeline; humans retain release decisions.

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11. Conclusion

NAEOS offers a structural answer to modern software engineering fragmentation: a platform that enforces **the specification as the source of truth**, **the constitution as governing law**, **NEIR as the unified model**, and **AI as a curated partner** — not a replacement for human judgment.

Under Apache License 2.0, with a vendor-neutral architecture and a growing ecosystem, NAEOS invites organizations and communities to build a more disciplined, traceable, and reproducible future of engineering — where you describe your system **once**, and build it **anywhere**.

NAEOS OSS — "Engineering With Discipline"

This document is based on the actual state of the project (NAEOS-foundation/naeos repository; software release baseline v3.7.1; engineering milestone P1.11) and is intended for publication, technical evaluation, and adoption discussions. All technical claims can be verified in the official project documentation (docs/NES-, specification/, constitution/).*